

Master Thesis Guidelines

Agenda

- Introduction
- Master thesis rules
- Tips, notes and useful links.

Cooperation plan

- Presentation of the proposal of the topic of the thesis and previous experience (**ready** – when registering for the seminar).
- Specification of the research problem, literature review, detailed work plan – **mid-December**, 1 week for feedback.
- Proof of work (initial version of the model, implemented in the learning environment, etc.) – **end of the first semester**, 1 week for feedback.
- Ready model with preliminary analysis of results – **mid-June**, 2 weeks for feedback.
- Finished text – **end of August**, 2 weeks for feedback.
- Work submitted for acceptance – **two weeks before the university deadline**, a week for my comments
- Submitting a thesis

Cooperation model

- The files must be sent by e-mail in .docx or .pdf format (if the work is prepared in LaTeX).
- In addition to the title of the thesis, the file name should contain the author's name, surname and index number.
- I should have access to all the source codes:
- If you are writing a paper in LaTeX, I suggest using [Overleaf](#)
- The code should be stored in the appropriate [GitHub](#) repository
- The data should be made available in an appropriate repository. If the file is too large, then on the SGH Cloud.
- If you use the sensitive data, it is necessary to obtain appropriate consents and proper [anonymization of the data](#).
- After implementing my remarks, do not delete my comments, but mark them as done.

Plan of thesis

- By mid-December, you should prepare a plan of your thesis. The document should include:
 - Problem description
 - Economic significance of the problem
 - Research hypotheses
 - Research methods, including the description of the data you will use
 - Table of contents of your final thesis
- The concept should be a short document, about 2 pages and should be sent to me by e-mail in the form of a .docx or .pdf file

Plan of thesis

- What should be included in the description of the problem?
 - owner (who has this problem)
 - goal (it must be known what this owner wants to achieve)
 - deviation of the actual state from the expected one (i.e., the level of satisfaction of the goal must be insufficient)
- Economic importance:
 - How are the results related to the theory of economics?
 - What real economic problems does your thesis solve?
 - Economic importance of a given sector of the economy (e.g., the cost of undiagnosed diseases as a motivation to build a disease-detecting model)

Plan of thesis

- You should choose a topic for your thesis on your own:
 - However, I can help with clarifying it and stating the hypotheses!
- The thesis plan should also briefly describe a data source you intend to use.
- It is possible to continue the cooperation, e.g., during the doctorate at SGH Warsaw School of Economics.

Formal Rules

- Masters' thesis:
 - Dealines
 - Formal requirements

Thesis Outline

- Title page (according to the dean's office template)
- Table of contents
- Introduction
- First chapter:
 - "soft" theoretical chapter
 - Definition of the purpose of work, its practical significance, in particular in economically related fields
 - literature review in the subject of the thesis
 - Formulation of research hypotheses and their justification

Thesis Outline

- Chapter two:
 - Data description and exploratory data analysis,
 - The goal is to show that the available data allows to solve the problem
- Chapter three:
 - “strong” theoretical chapter. A detailed description of the analytical tools and techniques used in the work, indicating that the selected methods are adapted to the available data and allow for the verification of the hypotheses.
- Chapter four:
 - Experimental chapter. Description of models or tools used in a thesis, description of the training process and sensitivity analysis of the results (how the model reacts to the change of initial parameters, hyperparameter tuning)
 - Verification of the hypotheses based on the model results; detailed conclusions.

Thesis Outline

- Chapter five:
 - Conclusions and discussion.
 - Discussion of the limitations of the study
 - Discussion of how the results of the work can be applied in business practice
 - Discussion of possible directions for expanding research in the future
- Summary:
 - A paragraph or two that synthetically describes the most important results
- Bibliography
- List of figures and list of tables
- Attachments (including source codes for analyses – link to the GitHub repository)
- Abstract (approx. 900 words)

Typesetting

- Basic formal requirements:

- 1) A4 format, double-sided print;
- 2) margins: top - 2.5 cm, bottom - 2.5 cm, side - mirror image: inside - 3.0 cm, outside - 2.0 cm;
- 3) justified (text aligned along both side margins);
- 4) Times New Roman font;
- 5) text font size 12 points;
- 6) source footnote font size 10 points;
- 7) spaces between lines 1.5 line;
- 8) continuous page numbering throughout the work.

Typesetting

- I suggest using the university thesis template from the beginning!
- If you choose to use LaTeX, some useful links:
 - [Thesis Template](#)
 - Introduction to Overleaf [pt. 1](#) and [pt. 2](#)
 - [Introduction to Git with Overleaf](#)
 - [LaTeX: Common mistakes](#)
 - [Basics of plotting with LaTeX](#)
 - [LaTeX diagrams generator in Julia](#)
 - [LaTeX figures generator in Julia](#)

Bibliography

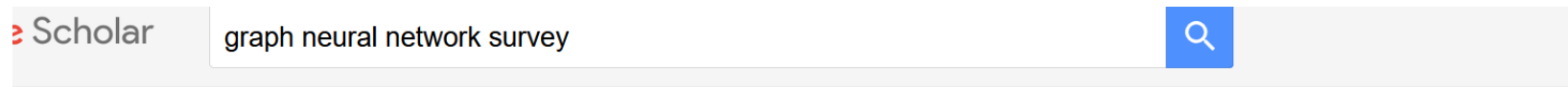
- Bibliography – singular at the end of the work, without division into types, sorted alphabetically, only items cited directly in the body of work
- According to the university's guidelines, 2 standards are acceptable:
 - 1) the Harvard system, i.e. footnotes inside the text, e.g. (Kowalski 2006, p. 31-52), or in the list of literature e.g. 1. Mitra, A., 1998. Fundamentals of Quality Control and Improvement, New Jersey, Prentice Hall; footnotes used only for explanations; or
 - 2) a system recommended by the Publishing House of the SGH Warsaw School of Economics, i.e. footnotes at the bottom of the page with continuous numbering e.g. 1A. Mitra, Fundamentals of Quality Control and Improvement, Prentice Hall, New Jersey, 1998, p. 28, in the list of literature e.g. 1. Mitra A., Fundamentals of Quality Control and Improvement, Prentice Hall, New Jersey, 1998.

Bibliography

- When quoting a bibliographic item, we write about what is important for the thesis, not everything that is discussed in the cited work.
- Remember that the diploma thesis tests your knowledge and understanding of the tools you use. In the case of key publications (e.g. introducing the model you use), the link alone is not enough – you need to describe the tools in detail!
- When using Internet resources (reports, websites, etc.), the bibliography should include a link and the date of access
- It is not allowed to refer to "obvious truths" and "common sense": all claims should be supported by literature and correctly stated, for example, as follows: "Drummond (2000) proves that ..."
- When looking for publications, remember to use [SGH VPN](#)!

How to look for the literature

- The first strategy is to look for review articles:
 - We enter the term we are interested in in a search engine (e.g. Google Scholar or Web of Science), e.g. *graph neural networks* with an additional keyword: **survey**, **challenges**, **methods of**
 - Examples:



[A survey on graph neural networks](#)

[\[PDF\] eai.eu](#)

J Wang - EAI Endorsed Transactions on e-Learning, 2022 - publications.eai.eu

... **graphs** by passing messages between **graph** nodes. This paper introduces commonly used **graph neural networks**, ... for **graph neural networks**. It also provides an outlook on the future of ...

☆ Zapisz Cytuj Cytowane przez 17 Powiązane artykuły Wszystkie wersje 2

[Challenges in sentiment analysis](#)

[\[PDF\] cuni.cz](#)

SM Mohammad - A practical guide to **sentiment analysis**, 2017 - Springer

... some of the **challenges** that still remain. We start by discussing different **sentiment analysis** problems and how one of the **challenges** is to explore new **sentiment analysis** problems that ...

☆ Zapisz Cytuj Cytowane przez 123 Powiązane artykuły Wszystkie wersje 11

How to look for the literature

- Then, we read such an article (we choose those that have many citations and are relatively up-to-date; a literature review from 24 years ago with 10 citations is not important!).
- Then we look at a bibliography of such an article, and we choose the articles we are interested in.

How to look for the literature

- Second approach: we found a paper that interests us, and then we try to find similar publications.
- We look at its bibliography and select the interesting publications.
- In the next step, we look at who cited this paper at Google Scholar or Web of Science:

Graph neural networks designed for different **graph** types: A survey

[PDF] arxiv

[JM Thomas, A Moallem-Oureh...](#) - arXiv preprint arXiv ..., 2022 - arxiv.org

... the **graph** types is Zhou et al. (2020). However, it does not consider all **graph** types covered in this **survey** since we provide a more fine-grained distinction of different **graph** properties. ...

☆ Zapisz  Cytuj **Cytowane przez 35** Powiązane artykuły Wszystkie wersje 6 

How to look for the literature

- And from this list, we select potentially interesting publications:

Bez ograniczenia
czasowego

Od 2024

Od 2023

Od 2020

Zakres
niestandardowy...

Wg trafności

Wg daty

Dowolny język

Tylko język polski

✉ Utwórz alert

Graph neural networks designed for different graph types: A survey

☐ Szukaj w artykułach zawierających cytaty

Graph neural networks for temporal graphs: State of the art, open challenges, and opportunities

[A Longa](#), [V Lachi](#), [G Santin](#), [M Bianchini](#), [B Lepri](#)... - arXiv preprint arXiv ..., 2023 - arxiv.org

Graph Neural Networks (GNNs) have become the leading paradigm for learning on (static) graph-structured data. However, many real-world systems are dynamic in nature, since the ...

☆ Zapisz Cytuj Cytowane przez 56 Powiązane artykuły Wszystkie wersje 12 >>

[PDF] arxiv.org

Towards graph foundation models: A survey and beyond

[J Liu](#), [C Yang](#), [Z Lu](#), [J Chen](#), [Y Li](#), [M Zhang](#)... - arXiv preprint arXiv ..., 2023 - arxiv.org

Emerging as fundamental building blocks for diverse artificial intelligence applications, foundation models have achieved notable success across natural language processing and ...

☆ Zapisz Cytuj Cytowane przez 66 Powiązane artykuły Wszystkie wersje 4 >>

[PDF] arxiv.org

Graph neural networks for text classification: A survey

[K Wang](#), [Y Ding](#), [SC Han](#) - Artificial Intelligence Review, 2024 - Springer

[PDF] springer.com

How to look for the literature

- During your query, remember that it is important to pay attention to the date of publication and the number of citations. The rule of thumb is that we are interested in publications that have plenty of citations or those that are quite recent!
- At the first step, do not read the entire work; start with an abstract, and if the paper fits your topic, then go through the main content of the work.
- When working with books or textbooks, read only the chapters that are important to your thesis, not the whole book.

Numbering

- We number: chapters, tables, drawings, formulas
- We always use Arabic numerals
- The numbers of tables, drawings, and formulas are consecutive throughout the work: 1, 2, 3, ... (without prefixes with chapter number)
- Chapters and subchapters are numbered: 1, 1.1, 1.2, 2, 2.1, ..., e.g. "5. Final conclusions"
- The formulas should be centred, and the numbers should be given in brackets on the right, e.g.:

$$2 \int_{-1}^1 \sqrt{1-x^2} dx \neq \lim_{n \rightarrow +\infty} \exp \left(\pi^2 + \frac{1}{n} \right) \quad (1)$$

- We always refer to the number of the table, equation, etc., in the text.

Tables and Figures

- The descriptions of tables or figures must include:
 - Number
 - A detailed description of what the drawing depicts
 - The set of parameters used to generate it
 - The caption under the table or figure must say what conclusion follows from it
 - Entry about the source (if not generated on your own; in *italics*, aligned to the right, above the description)
- Each table or figure must be used in the main text, and each must also have a description. A description must be written in a way that the reader will understand what happens in the figure without reading the main body of text!

Tables and Figures

- Example:

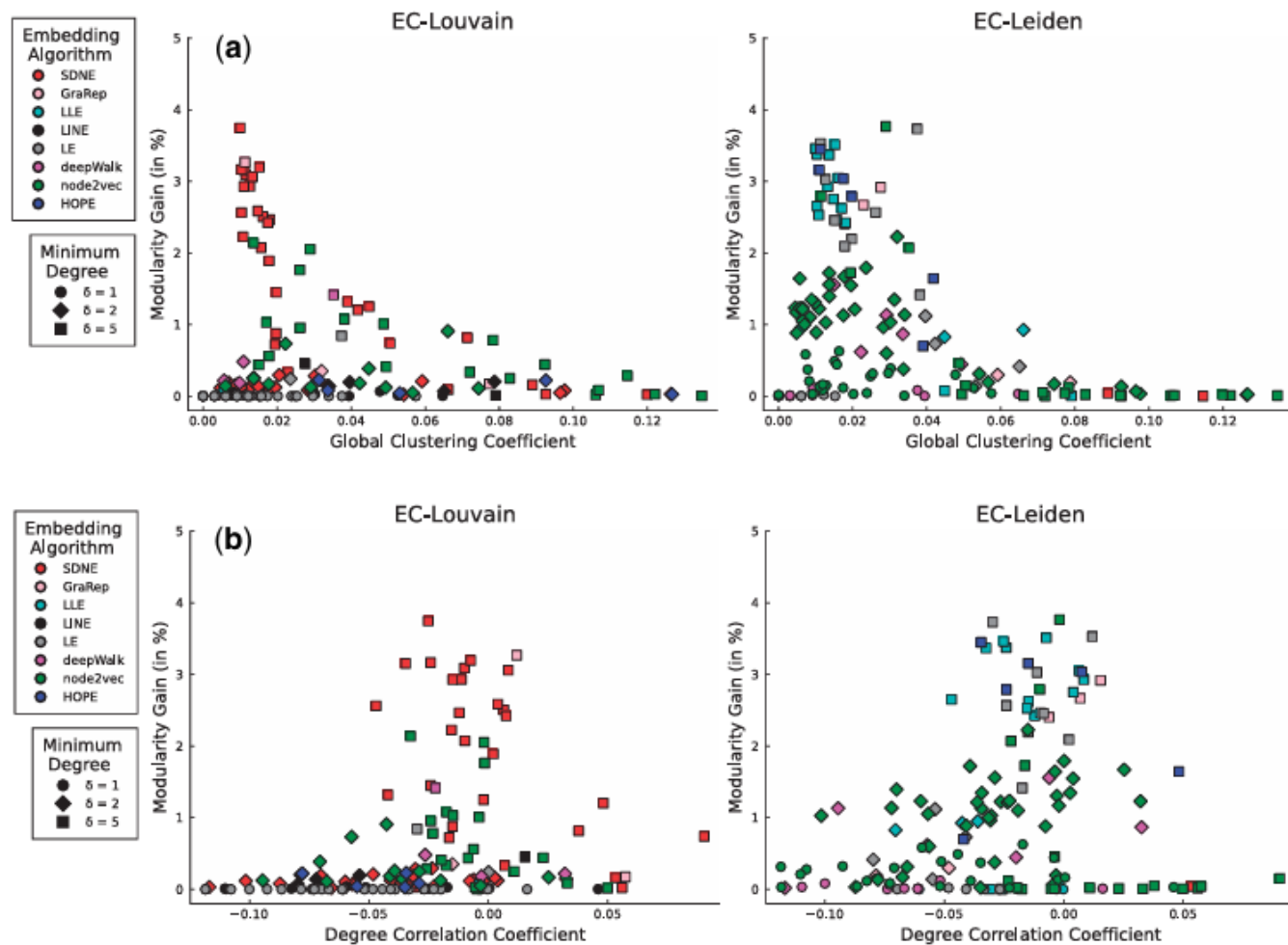


Fig. 4. Modularity increase (in percentage) from various embedding-clustering combinations as a function of (a) global clustering coefficient and (b) correlation coefficient. EC-Louvain is plotted in the left column, EC-Leiden in the right.

Introduction

- Usually, a Thesis starts with the sentence: "The purpose of the work is..."
- Then, we briefly describe the research problem
- Hypotheses emerge from the description of the problem
- Hypotheses lead to research methods
- The research methods impact the structure of the work

First Chapter

- At this point, we move on to a more detailed description of the research problem
- A brief history of research on the given issue should be presented
- Similarly, it is a good idea to present a description of research on the related problems
- We describe the economic importance of the thesis
- We describe in detail the hypotheses and how they relate to the main goal of the thesis

Second Chapter (Data Analysis)

- We discuss the source of the data or how the data was collected
- We introduce the notation
- We carry out an exploratory analysis of the data
- We interpret its results, in particular, how they influence the choice of a modelling technique in the thesis

Chapter Three (Theoretical Chapter)

- We describe in detail the experiments carried out in the thesis
 - What do we want to do? What steps will we take?
- We describe selected models in detail
 - Remember to explain the theoretical foundations of the models, use equations and describe each of them
 - Explain why the selected model(s) are best tools for verifying the hypotheses
- We introduce the implementation details
 - programming language, pipelines and other tools, hardware used for computations
 - If you need more computational power (e.g. Amazon Cloud or supercomputer access), please let me know

Chapter Four (Experiments)

- We describe the model training process and the overall experiment
- We tune the hyperparameters and/or perform a sensitivity analysis of the model – check how it reacts to various combinations of parameters
 - Alternatively, we compare different classes of models to find the most efficient ones
- We analyze the results:
 - We describe the parametrization of the final model – this is important as a form of validation; it allows us to show that the model works and returns reasonable results
 - We describe metrics used to verify the stated hypotheses

Final Chapters

- In the discussion/conclusions chapter:
 - We describe whether the results obtained validate or negate the research hypotheses
 - Rejecting the initial hypotheses is also a valid and worthwhile result!
 - We avoid the “maybe” statements! If the results confirm the hypotheses, then the conclusions are valid and unambiguous
 - We discuss the limitations of the conducted study and possible future directions
 - We discuss how the results of the work can be applied in business practice
- In summary:
 - We summarize the most important results in 2-3 paragraphs, so that the reader can quickly understand what the most important result of the work is.

Language

- The text should be written in a stylistically consistent manner (one font type and font size for text, one for chapter titles, etc.)
- Avoid using colors and bold text (apart from mathematical notation, e.g. vectors) – if you want to highlight a fragment of the text, use *italics*
- To write down the code, use a monospace font, e.g. Consolas

Language

- Avoid colloquial language and metaphors
- The thesis is a professional document; write it with simple and precise language
- Do not use imprecise terms such as "certain", "many", "suitable", "some", ... All conclusions drawn should be unambiguous (if you cannot draw an unambiguous conclusion from the results, it means that it is not justified and should not be described in the thesis!).
- The conclusions should not be general; a statement, "the results obtained confirm all the hypotheses", is wrong
- Chapters should not be too short (1-1.5 pages) or too long (20-30 pages)

Language

- The work should flow logically and should be a uniform block of text: each paragraph and each sentence of the work should be related to the previous one.
- Each fragment of the main text must refer to the main topic. If you need to describe something unrelated yet important (e.g. specific implementation details), use footnotes.
- Similarly, the subsequent chapters should stem from the previous ones.
- We write only about facts, not about what we think or what visions we have; All facts should be documented, whether by literature or results
- We write what we did, not what we didn't do.

Summary

- Remember that by choosing my seminar, you consent to the processing and storing of your personal data in Decision Support and Analysis Unit
- The principles described in this document are merely a suggestion: there are other ways of writing a good thesis. However, they allow you to speed up the process significantly.
- Before you send me the thesis for marking, read it again. Sometimes it's worth leaving it for a few days and coming back to it later with a "fresh eye".
- When I send you a text with comments, please do not remove them after solving the issues – rather mark them as done. This will significantly speed up the proofreading process.

Summary

- Remember that you can contact me at any time. I will always try to help you, but this is your thesis, and its final quality will depend on you and your workflow.